

The Open Methodology MoBI Laboratory

Nicole Burke, Nathalia Bianchini Esper, Bryan Gonzalez, Apurva Gokhe, Alp Erkent, Iktae Kim, Camilla Strauss, Celia Maiorano, Freymon Perez, John Vito d’Antonio-Bertagnolli, Stan Colcombe, Alexandre Franco, Michelle Freund, Michael P. Milham
Child Mind Institute, New York

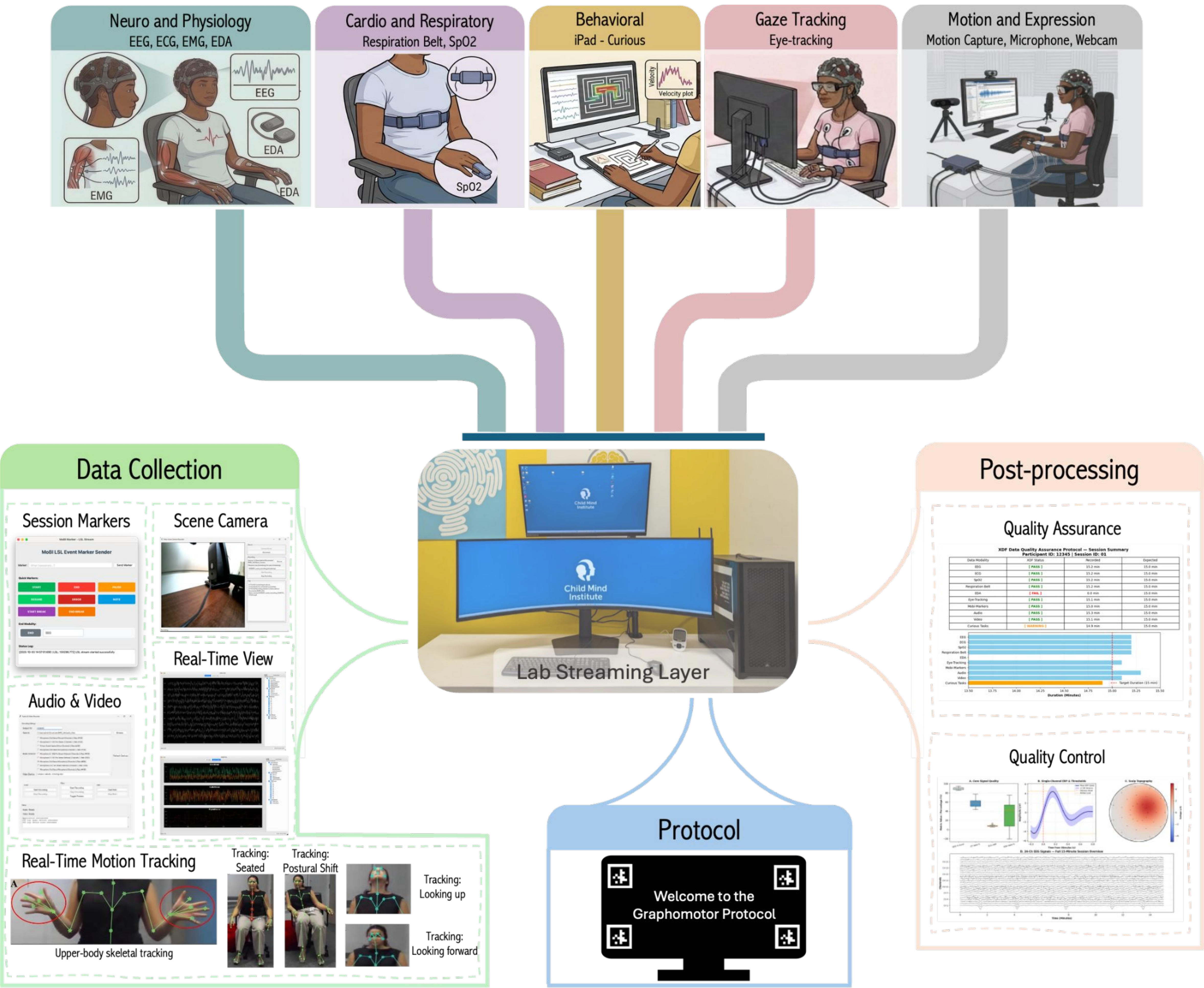


Introduction

The past 25 years has seen an explosion in **multimodal brain imaging studies**¹.
Integrating complementary data types provides richer insights into **brain function, structure, and connectivity**^{2,3}.

Traditional MoBI Bottleneck

- Heavy Hardware Complexity:** Managing 3–5 separate PC towers and displays.
- Burdensome Operations:** Manual multi-device start/stop sequences.
- Time Drift & Sync Hurdles:** Severe timestamp mismatches requiring tedious post-hoc adjustments.



- Unified Clock:** LSL uses a shared high-precision system clock for sub-millisecond real-time hardware alignment.
- No post-hoc processing:** Eliminates the time-consuming burden of stitching together separate log files and synchronizing timestamps offline.
- No post-hoc time syncing:** Eliminates offline timestamp alignment and manual cross-device time drift correction.
- Automated QA and QC:** Open-source batch processing tool enables instant automated checks for protocol length, stream continuity, and signal integrity.

Global Replication

- Two labs in USA**
Child Mind Institute
- One lab in South Africa**
Stellenbosch University
- Two labs in Brazil**
University of São Paulo
Federal University of Rio Grande do Sul

Impact

- 5x** Successful lab replications
- ~30%** Cost reduction
- 8** Open source tools
- 100%** Open source documentation



Contact information:
Nicole Burke - nicole.burke@childmind.org
Nathalia Esper - nathalia.esper@childmind.org
John Vito - john.vito@childmind.org

References:
[1] Warbrick, 2022
[2] Calhoun & Sui, 2016
[3] Wagner et al., 2019